



Semisupervised SAR image change detection based on a siamese variational autoencoder

Guangwei Zhao^{*}, Yaxin Peng

Department of Mathematics, School of Sciences, Shanghai University, Shanghai 200444, China

ARTICLE INFO

Keywords:

Synthetic aperture radar (SAR) images
Change detection
Variational autoencoder
Siamese structure
Semisupervised learning

ABSTRACT

In synthetic aperture radar (SAR) image change detection, the deep learning has attracted increasingly more attention because the difference images (DIs) of traditional unsupervised technology are vulnerable to speckle noise. However, most of the existing deep networks do not constrain the distributional characteristics of the hidden space, which may affect the feature representation performance. This paper proposes a variational autoencoder (VAE) network with the siamese structure to detect changes in SAR images. The VAE encodes the input as a probability distribution in the hidden space to obtain regular hidden layer features with a good representation ability. Furthermore, subnetworks with the same parameters and structure can extract the spatial consistency features of the original image, which is conducive to the subsequent classification. The proposed method includes three main steps. First, the training samples are selected based on the false labels generated by a clustering algorithm. Then, we train the proposed model with the semisupervised learning strategy, including unsupervised feature learning and supervised network fine-tuning. Finally, input the original data instead of the DIs in the trained network to obtain the change detection results. The experimental results on four real SAR datasets show the effectiveness and robustness of the proposed method.

1. Introduction

Remote sensing image change detection analyzes two or more preregistered images acquired in the same geographic area at different times and finds the changed areas or pixels (Gong, Zhan, Zhang, & Miao, 2017). It plays an important role in many fields, such as the location and disaster assessment of earthquake areas (Wang, Jiao, & Yang, 2016), the monitoring of crop growth conditions (Liu, Bruzzone, Bovolo, & Du, 2015), and land cover map updating (Fernandez-Prieto & Marconcini, 2011). Synthetic aperture radar (SAR) data have attracted increasingly more attention in the application of change detection because SAR is an active sensor independent of climate and light. Compared with ordinary optical images, SAR images can provide more information on ground objects. However, the inherent speckle noise makes SAR image change detection more challenging.

As mentioned in Bruzzone and Prieto (2002), unsupervised change detection mainly includes three steps: (1) image preprocessing, such as radiometric correction, image registration, and image denoising; (2) generating the difference image (DI) through the preprocessed data; and (3) classifying the DI and output change map. At present, the research mainly focuses on the latter two of the above three steps. The log ratio (LR) (Li, Martinis, Plank, & Ludwig, 2018; Zheng et al., 2017), mean ratio (Inglada & Mercier, 2007), and neighbor-based ratio (NR) (Gong, Cao, & Wu, 2012) operations are widely used for DI generation. Considering that a single operator is insufficient to describe the change information, different fusion strategies (Gong, Li, Jiao, Jia, & Su, 2014; Hou,

^{*} Corresponding author.

E-mail addresses: zhaoguangwei@shu.edu.cn (G. Zhao), yaxin.peng@shu.edu.cn (Y. Peng).

Wei, Zheng, & Wang, 2014; Ma, Gong, & Zhou, 2012) were used to improve the quality of DIs. The quality of a DI plays a decisive role in classification accuracy.

The threshold method and clustering method are commonly used in DI classification. The classic threshold method uses the expectation maximization (EM) algorithm (Bruzzone & Prieto, 2000) or Kittler–Illingworth (KI) algorithm (Kittler & Illingworth, 1986) to obtain the global threshold based on the construction of a statistical model. Considering that the KI algorithm cannot accurately distinguish between changed and unchanged categories, the KI threshold algorithm based on generalized Gaussian (GG) modeling was proposed in Bazi, Bruzzone, and Melgani (2005). Su, Gong, Sun, and Jiao (2014) proposed a semi-EM (SEM) algorithm based on the GG model. Although the threshold method is easy to understand, the speckle noise in a DI makes accurate modeling difficult. The clustering algorithm is an unsupervised classification method that does not need to be modeled. The fuzzy c-means (FCM) performs classification in a probabilistic manner and can retain more information than hard clustering (Ghosh, Mishra, & Ghosh, 2011; Hu, Zhong, Xie, Lv, & Lv, 2021). To enhance the robustness of the algorithm, Mishra, Ghosh, and Ghosh (2012) introduced spatial contextual information into the FCM and proposed the FCM algorithm with local information. Liu, Wang, Huang, Li, and Jiao (2020) proposed the ensemble strategy multiobjective fuzzy clustering method (ESMOFCM), which alleviates the contradiction between image denoising and detail preservation. Although clustering methods achieve good classification results, they are susceptible to speckle noise. In addition, a clustering algorithm relies on the initial clustering center, which means that the algorithm can easily fall into the local optimum.

The traditional process of generating DIs results in information loss, and the improper selection of the threshold and clustering method will make the change detection result unstable (Liu, Gong, Qin, & Zhang, 2018). The development of artificial intelligence brings new ideas to solve these problems. Benefiting from the ability of deep learning to automatically extract multilevel image features, the research of change detection based on deep networks is in full swing (Shi, Zhang, Zhang, Chen, & Zhao, 2020). Planinsic and Gleich (2018) employed a stacked autoencoder network to classify the SAR image features extracted by an adjustable Q discrete wavelet transform. Chen et al. (2019) proposed a fast unsupervised change detection method that used a stacked autoencoder to generate DIs in an unsupervised manner and a fuzzy clustering algorithm to achieve classification. Because convolutional neural networks (CNNs) have strong deep feature learning capabilities, they have achieved satisfactory results in image processing tasks. Liu, Li, Cao, and Shen (2017) designed a dual-channel CNN (DCCNN) with supervised learning for SAR image change detection. In order to avoid the loss of information in the DI generation process, Li et al. (2019) used a CNN to directly extract features from the original images for classification. Gao, Wang, Gao, Dong, and Wang (2019) proposed a change detection method based on convolution-wavelet neural networks in 2019, which introduces dual tree complex wavelet transform into CNN to reduce the impact of speckle noise. In view of the limited expression ability of spatial domain features and patch-based feature analysis being affected by noise at the boundary, Qu, Gao, Dong, Du, and Li (2021) presented a dual-domain CNN including the spatial domain and frequency domain for SAR image change detection. Compared with traditional methods, these learning-based algorithms are more effective and robust. However, they lack the mathematical description and regularization of potential spatial distribution features, which may weaken the feature representation ability of networks.

A variational autoencoder (VAE) (Kingma & Welling, 2014) is a generative model proposed by Kingma et al. in 2014. Compared with the autoencoder (AE) (Hinton & Salakhutdinov, 2006), a VAE reduces the data dimensionality and can describe the potential space using probability. By learning the probability distribution of each potential attribute, a VAE can obtain the continuous space to which a feature belongs, so it has good feature representation capabilities. Since the VAE was proposed, it has been applied to tasks such as image clustering (Prasad, Das, & Bhowmick, 2020), medical image registration (Dalca, Balakrishnan, Guttag, & Sabuncu, 2018), and point cloud segmentation (Meng, Gao, Lai, & Manocha, 2019). To the best of our knowledge, there are relatively few studies on using the VAE for SAR image change detection. In addition, most of the change detection methods based on a deep network framework send the corresponding two samples into the network in series or parallel. Because the bitemporal images are fused as input before being fed into the networks, the networks fail to provide the depth feature information of individual raw images to help classification (Zhang et al., 2020). The siamese structure (Bromley et al., 1993) with the same structure and parameter sharing subnetwork can extract individual image information separately and maintain the spatial consistency of the extracted features, which is beneficial to identifying the different information between images.

Inspired by the above, this paper proposes semisupervised SAR image change detection based on the siamese variational autoencoder. The algorithm includes the following three aspects: (1) preclassification is used to obtain some high-precision label data, (2) the deep VAE network is constructed to learn the image features and then fine-tune the neural network parameters, and (3) the trained deep neural network is used to classify the changed pixels and unchanged pixels.

The main contributions of this paper include:

- (1) This paper introduced the VAE network to the SAR image change detection task for feature learning. The continuity and integrity of the hidden space are ensured by constraining the spatial features to obey a specific distribution.
- (2) We design a siamese variational autoencoder network to ensure the feature spatial consistency of the bitemporal images, which is helpful to recognizing the changed and unchanged information between images.
- (3) Semisupervised learning can achieve good classification results with limited labeled data. In addition, greedy layerwise training can better initialize the network parameters.
- (4) Directly classify the original image to avoid the information loss caused by DI generation and the unstable change detection result generated by the traditional method.

The subsequent content is organized as follows. In Section 2, we review the principles of the VAE. Section 3 describes the proposed network framework and the implementation process in the SAR images change detection task in detail. The dataset description and evaluation criteria are given in Section 4. Then, the parameter analysis and contrast experiments are given in Section 5. In the final section, the conclusion is presented.

2. Principle of the variational autoencoder

The VAE is composed of generative model $p_\theta(x) = p_\theta(z)p_\theta(x|z)$ and inference model $q_\phi(z|x) \rightarrow p_\theta(z|x)$. The purpose is to solve the maximum likelihood estimation of $\log p_\theta(x)$:

$$\log p_\theta(x) = L(\theta, \phi; x) + D_{KL}(q_\phi(z|x)||p_\theta(z|x)) \quad (1)$$

where x is the observed data, and z is the latent variable. θ and ϕ are the parameter sets that need to be optimized. The first term is the variational lower bound, which is also called the evidence lower bound (ELBO). The second term $D_{KL}(p(A)||p(B))$ represents the Kullback–Leibler (KL) divergence of the distribution of A and B :

$$\begin{aligned} D_{KL}(q_\phi(z|x)||p_\theta(z|x)) &= E_{q_\phi(z|x)} \log \frac{q_\phi(z|x)}{p_\theta(z|x)} \\ &= E_{q_\phi(z|x)} [\log q_\phi(z|x) - \log p_\theta(z, x)] + \log p_\theta(x) \end{aligned} \quad (2)$$

By Eqs. (1) and (2), we can obtain:

$$\begin{aligned} L(\theta, \phi; x) &= -E_{q_\phi(z|x)} [\log q_\phi(z|x) - \log p_\theta(z, x)] \\ &= -E_{q_\phi(z|x)} [\log q_\phi(z|x) - \log(p_\theta(z) \cdot p_\theta(x|z))] \\ &= E_{q_\phi(z|x)} [\log p_\theta(x|z)] - D_{KL}(q_\phi(z|x)||p_\theta(z)) \end{aligned} \quad (3)$$

The larger the first term in Eq. (3) is, the closer the sample reconstructed by resampling z is to the original sample x . That is, the reconstruction error is smaller. In the above formulas, $p_\theta(z|x)$ represents the true posterior distribution, $q_\phi(z|x)$ represents the distribution generated by the inference process, and $p_\theta(z)$ is the prior distribution.

Since the KL divergences are nonnegative, we can acquire the following from Eqs. (1) and (3):

$$\max_{\theta, \phi} \log p_\theta(x) \propto \max_{\theta, \phi} L(\theta, \phi; x) \propto \min_{\theta, \phi} (D_{KL}(q_\phi(z|x)||p_\theta(z)) - E_{q_\phi(z|x)} [\log p_\theta(x|z)]) \quad (4)$$

where $D_{KL}(q_\phi(z|x)||p_\theta(z))$ represents the regularization term, and $-E_{q_\phi(z|x)} [\log p_\theta(x|z)]$ is the reconstruction error. θ and ϕ are parameter sets.

3. Methodology

Consider the preregistered SAR images I_1 and I_2 , which are captured from the same area at times T_1 and T_2 , respectively. The purpose of change detection is to determine whether each pixel in two images has changed, which is essentially a classification problem. A binary map I with the same size as I_1 and I_2 is generated by the classification algorithm, where 0 and 1 represent the unchanged category and changed category, respectively.

The framework of the traditional method is shown in Fig. 1. As mentioned in Section 1, the change detection accuracy will decrease when a DI affected by speckle noise.

Selecting the appropriate clustering or threshold method is not easy, and selecting the wrong method will cause the change detection results to be unstable.

The change detection process proposed in this paper is shown in Fig. 2. Our method extracts features from the original images and generates classification results directly, which avoids the shortcomings of traditional methods. This section will introduce the selection of training samples, the proposed network structure, and the semisupervised learning process.

3.1. Preclassification and sample selection

Because the FCM does not consider spatial information in DI classification, the results are seriously affected by speckle noise. If an excellent sample selection strategy is not used, the training set will contain many low-quality samples. Therefore, we need to use FCM preclassification marks to select some “high-quality” pixels to train our network. This paper uses the sample selection method in Shang, Yuan, Jiao, Meng, and Ghalamzan (2018). Using dependable samples, where dependable means that the training samples are accurate and diverse, to train a classifier can obtain excellent results. Considering the spatial continuity of an image, the number of the same labels in the neighborhood of each pixel label should be greater. We select a training set that contains pixels that satisfy the conditions shown in Fig. 3.

Each circle in Fig. 3 represents a pixel in the pseudolabel that belongs to the changed or unchanged category, and the triangle represents a pixel in another category. Figs. 3(a)–(c) show that the category of the central pixel is the same as that of most of the surrounding pixels, which conforms to the spatial continuity of the image and is selected as the training sample center. In contrast, if a central pixel class is different from most classes in the neighborhood, it may be considered noise and discarded, as shown in Figs. 3(d)–(f).

The sampling steps are as follows:

- 1: Count the number of pixels in the neighborhood whose label is the same as that of the center pixel, and record the number as n_i .
- 2: Using the formula $\frac{n_i}{n \times n}$ (n is the neighborhood size), we calculate the ratio of the number of same pixels to the total pixels in a neighborhood.

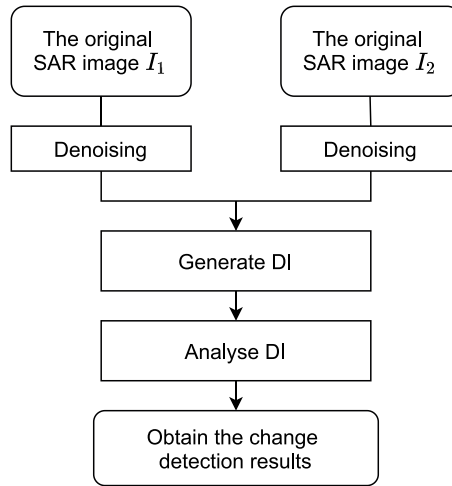


Fig. 1. The process of traditional change detection.

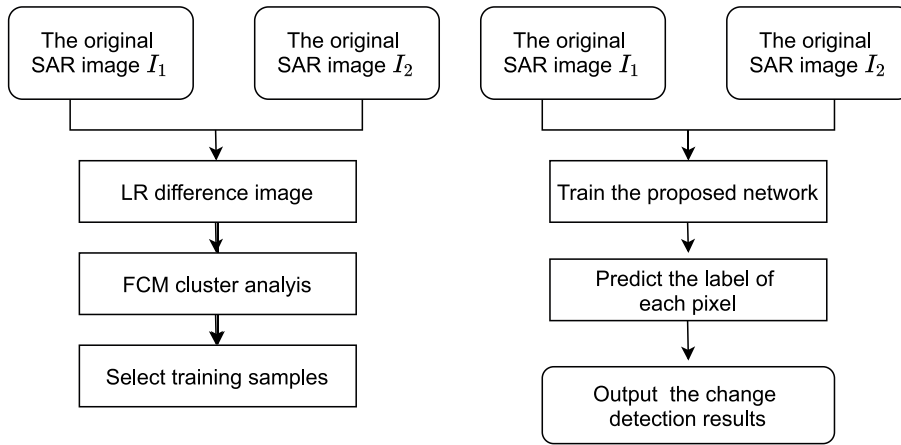
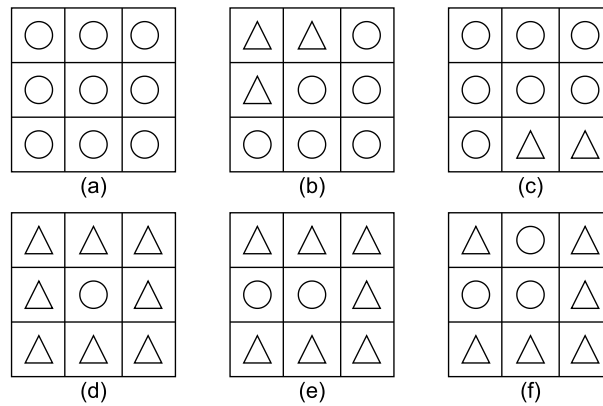


Fig. 2. Change detection process based on deep learning.

Fig. 3. Different cases in the 3×3 neighborhood of pixel i after preclassification.

3: If the ratio is greater than the set threshold α , the original image position corresponding to the center is used as the selected sample center.

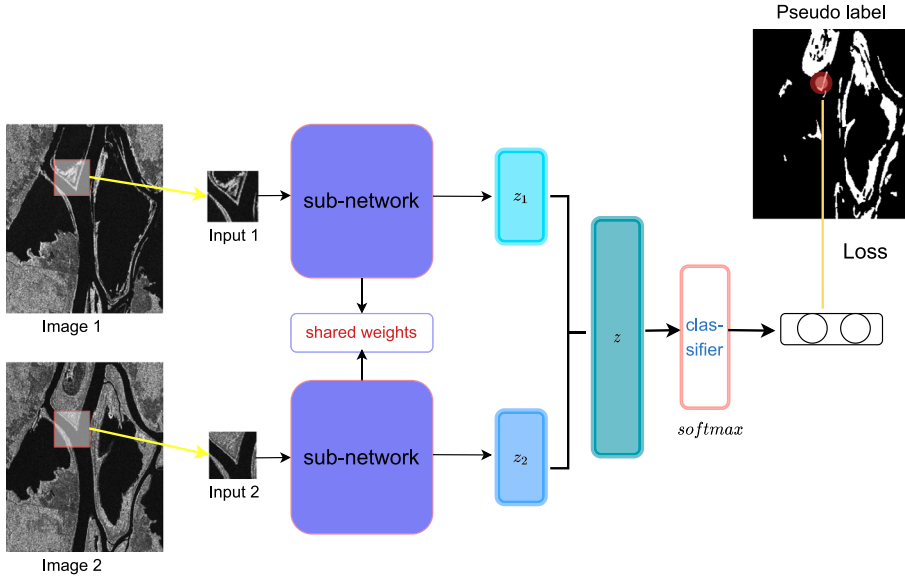


Fig. 4. The workflow of the proposed method.

The smaller or larger value of α is not appropriate. Smaller α is easy to cause incorrect samples to be selected. Larger α ensures the accuracy of samples but will sacrifice diversity. The value of α in this paper is 0.75.

3.2. Siamese variational autoencoder

The framework of the SAR image change detection algorithm proposed in this paper is shown in Fig. 4. The algorithm is composed of a coding layer, information fusion layer, and classification layer. Suppose that $I_1 = \{x_1(i, j); 1 \leq i \leq h, 1 \leq j \leq w\}$ and $I_2 = \{x_2(i, j); 1 \leq i \leq h, 1 \leq j \leq w\}$ are the bitemporal SAR images that need to be detected. $x(i, j)$ represents the gray value at position (i, j) . The network's input is the column vector drawn by the patch of the corresponding position in the two images, recorded as P_1 and P_2 , respectively. Two subnetworks with the same weight and structure are denoted as nonlinear mapping f . The forward propagation is as follows.

The primary function of the coding layer is to learn the nonlinear features of bitemporal images through multilevel networks. Shallow networks learn the primary features, and deep networks describe the regular distribution features of images:

$$\begin{cases} f_{\theta}(P_1) = z_1 \\ f_{\theta}(P_2) = z_2 \end{cases} \quad (5)$$

The above formula shows that the subnetwork with the same structure and shared parameters will output the image features with the same spatial structure and produce different corresponding responses to the changed samples.

The information fusion layer coalesces the features extracted from the two original images to obtain the differences between the data, similar to DI generation in traditional methods:

$$z = z_1 \oplus z_2 \quad (6)$$

The classification layer aims to analyze the difference information and map the difference information to the probability belonging to a specific category through a nonlinear classifier:

$$\text{Softmax}(Wz + b) = p \quad (7)$$

In the above formula, $\oplus$ represents the concatenate operation. θ , and $\{W, b\}$ are the parameters to be trained. The activation function used in the network is $\text{ReLU}(\cdot)$ (rectified linear unit, ReLU) (Glorot, Antoine, & Yoshua, 2011).

3.3. Subnetwork

The subnetwork is a deep variational autoencoder. As shown in Fig. 5, the network comprises an encoder (i.e. inference model) and a decoder (i.e. generative model), which perform feature learning and image reconstruction, respectively.

Take a dataset $x = \{x^i\}_{i=1}^n$, where x^i is assumed to be independent and identically distributed (i.i.d.) samples. For each input sample x^i ($\dim(x^i) = m$), a reconstruction sample $\hat{x}^i$ with the uniform dimension of the input is obtained through a series of nonlinear

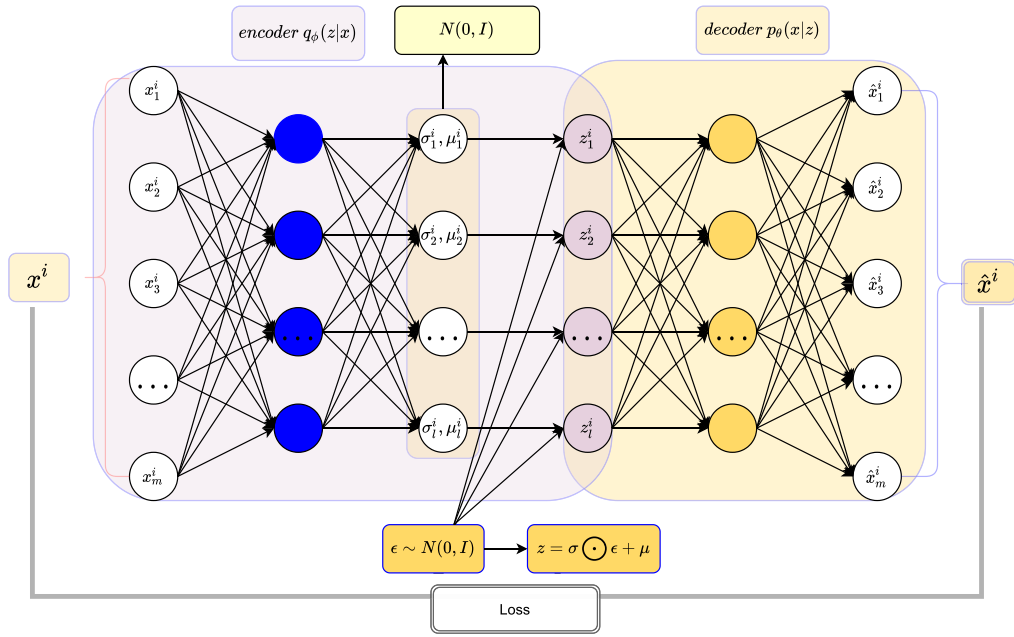


Fig. 5. VAE network structure diagram.

mappings. Based on the VAE principle in Section 2, we assume $p_\theta(z) \sim N(z; 0, I)$, $q_\phi(z|x^i) \sim N(z; \mu^i, \sigma^{2(i)}I)$, $p(\epsilon) \sim N(\epsilon; 0, I)$. The encoder and decoder are denoted as f and g , respectively.

For each piece of input data, a set of distribution parameters is obtained by a multilayer perceptron:

$$\begin{cases} f_{\theta_1}(x^i) = \mu^i \\ f_{\theta_2}(x^i) = \sigma^i \end{cases} \quad (8)$$

If the hidden variable z^i is obtained by sampling directly according to the distribution parameters, the backpropagation algorithm cannot be conducted. To solve the above problem, the variable ϵ , which obeys the standard normal distribution, is introduced for resampling:

$$z^i = \mu^i + \sigma^i \odot \epsilon \quad (9)$$

where $\odot$ is the Hadamard product.

Then, the output is:

$$g_{\theta_3}(z^i) = \hat{x}^i \quad (10)$$

According to the input and output data, the reconstruction error is obtained:

$$loss_{MSE} = \frac{1}{2} \|x^i - \hat{x}^i\|_2^2 \quad (11)$$

In addition, the regularization term is:

$$\begin{aligned} D_{KL}(q_\phi(z|x) \| p_\theta(z)) &= D_{KL}(N(z; \mu^i, \sigma^{2(i)}I) \| N(z; 0, I)) \\ &= \frac{1}{2} \sum_{j=1}^l (-\log(\sigma_j^i)^2 + (\mu_j^i)^2 + (\sigma_j^i)^2 - 1) \end{aligned} \quad (12)$$

The final loss function is as follows:

$$Loss(\theta_1, \theta_2, \theta_3; x^i) = \frac{1}{2} \|x^i - \hat{x}^i\|_2^2 + \frac{1}{2} \sum_{j=1}^l (-\log(\sigma_j^i)^2 + (\mu_j^i)^2 + (\sigma_j^i)^2 - 1) \quad (13)$$

where $l = \dim(z^i)$. θ_1 , θ_2 , and θ_3 are the parameters to be learned.

The above representation shows that the VAE can learn the characteristics of a specific target and the distribution to which the target belongs. For change detection tasks, since similar targets belong to the same category, the VAE has better discriminative characteristics than the AE.

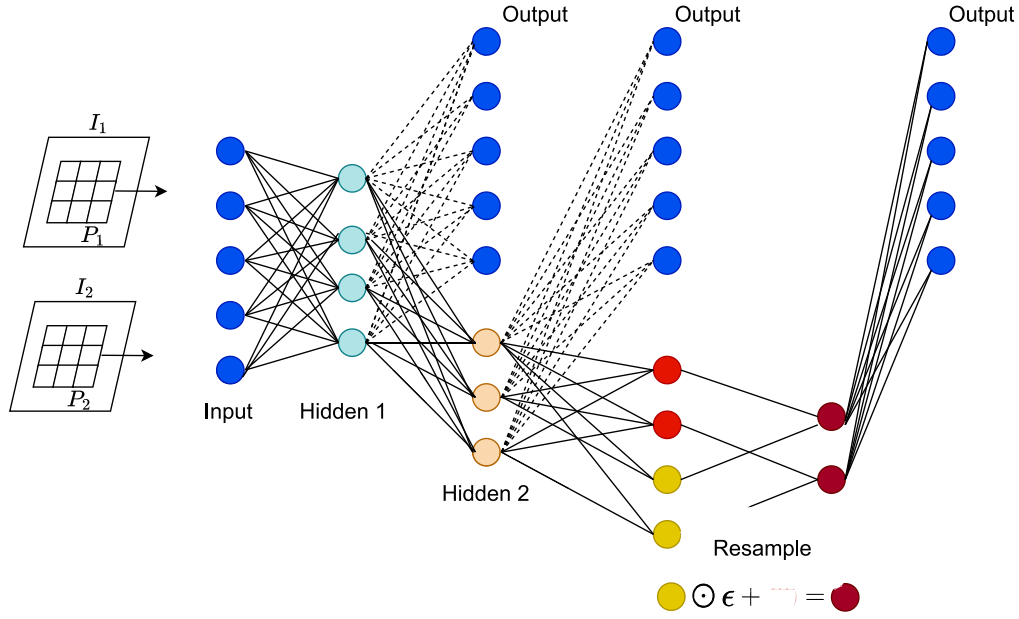


Fig. 6. The process of layer-wise training.

3.4. Semisupervised learning

In the SAR image change detection task, it is expensive to obtain a considerable amount of labeled data. In this case, it may not be easy to construct a reliable supervised classifier. The semisupervised classification method is especially suitable for the situation of a lack of labeled data (Van Engelen & Hoos, 2020). Extracting valuable features from unlabeled data or determining the initial parameters of supervised learning processes in an unsupervised way can help obtain a good classifier with a small number of labels (Li, Yang, Yang, Du, & Emery, 2020).

Nonlinear networks have multiple hidden layers, so it is challenging to optimize their weights and deviations. Random weights cannot always obtain satisfactory results. If the initial weight is large, the product usually cannot reach the global optimum. However, small initial weights lead to a small gradient in the early layer, making it impossible to train deeper networks.

Motivated by the algorithm in Gong, Zhao, Liu, Miao, and Jiao (2016), we first train the feature extractor layerwise in this paper. The learning process is shown in Fig. 6. P_1^i and P_2^i represent the i th input samples of the two subnets, and the output is the corresponding reconstructed image $\hat{P}_1^i, \hat{P}_2^i$. After the previous layer is trained, the learning of the next layer is conducted under the fixed parameters of the former layer. The reconstruction error function of the first two layers is as follows:

$$Loss_{\theta_1}(P_1^i, P_2^i) = \frac{1}{2} \|P_1^i - \hat{P}_1^i\|_2^2 + \frac{1}{2} \|P_2^i - \hat{P}_2^i\|_2^2 \quad (14)$$

$$Loss_{\theta_2}(P_1^i, P_2^i) = \frac{1}{2} \|P_1^i - \hat{P}_1^i\|_2^2 + \frac{1}{2} \|P_2^i - \hat{P}_2^i\|_2^2 \quad (15)$$

The loss function after resampling is as follows:

$$Loss_{\theta_3}(P_1^i, P_2^i) = Loss_{\theta_3}(P_1^i) + Loss_{\theta_3}(P_2^i) \quad (16)$$

$$Loss_{\theta_3}(P_1^i) = \frac{1}{2} \|P_1^i - \hat{P}_1^i\|_2^2 + \frac{1}{2} \sum_{j=1}^d (-\log(\sigma_j^i)^2 + (\mu_j^i)^2 + (\sigma_j^i)^2 - 1) \quad (17)$$

$$Loss_{\theta_3}(P_2^i) = \frac{1}{2} \|P_2^i - \hat{P}_2^i\|_2^2 + \frac{1}{2} \sum_{j=1}^d (-\log(\sigma_j^i)^2 + (\mu_j^i)^2 + (\sigma_j^i)^2 - 1) \quad (18)$$

$d = \dim(\sigma^i) = \dim(\mu^i)$, and θ represents the parameter set to be optimized.

Multiple pairs of unlabeled training samples were trained in turn. After training, the decoding layer is discarded, and the coding layer is retained.

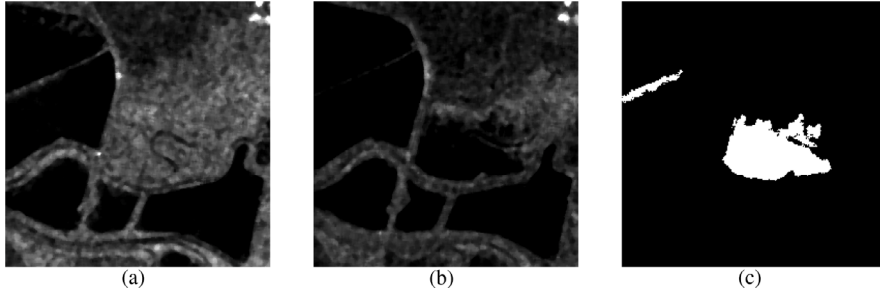


Fig. 7. Multitemporal SAR images of San Francisco. (a) image acquired in August 2003. (b) image acquired in May 2004. (c) ground truth.

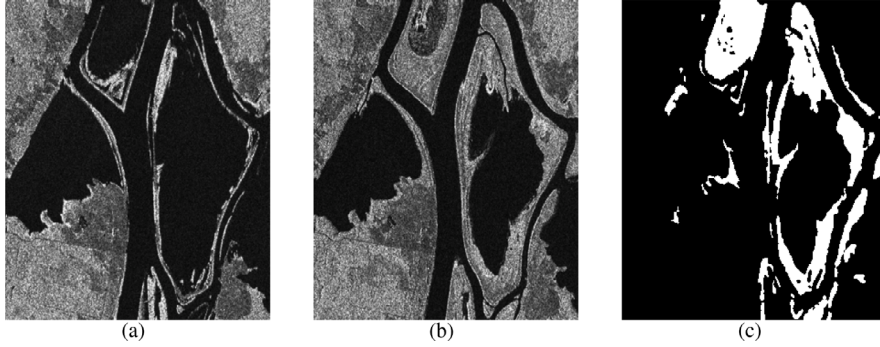


Fig. 8. Multitemporal SAR images of Ottawa. (a) image acquired in May 1997. (b) image acquired in August 1997. (c) ground truth.

The unsupervised feature extractor training completes the initialization of the network parameters. In addition, supervised network fine-tuning and classifier training are critical. In this process, high-quality label data selected by preclassification are used as prior information to guide the learning. As described in Section 3.2, the learning process of our method includes feature extraction, feature fusion, and output classification results. The entire network uses the cross-entropy error backpropagation strategy to fine-tune the weights to achieve the optimal classification. The cross-entropy loss is expressed as:

$$Loss = -[y^i \cdot \log(p^i) + (1 - y^i) \cdot \log(1 - p^i)] \quad (19)$$

where y^i is the sample label. The positive class is 1, and the negative class is 0. p^i represents the classification result of the prediction.

A deep siamese variational autoencoder network is established by training and fine-tuning. The neighborhood features of each position in the original image are input into the trained network, and the class labels of the pixels are output.

4. Dataset and evaluation criteria

4.1. Dataset

We conducted comparative experiments on three datasets. Each dataset contains three images. The first two images are real SAR images taken at different times on the same ground surface, and they are used to detect ground surface changes. The third image is the ground truth created by integrating prior information with photointerpretation based on the first two images.

The proposed network model is verified by the following three datasets. Before the experiment, all datasets were normalized to [0, 1].

San Francisco dataset: This dataset contains two 256×256 ERS-2 SAR images, which were obtained in August 2003 and May 2004, respectively, as shown in Fig. 7(a) and Fig. 7(b). The ground truth (i.e., reference image) is shown in Fig. 7(c).

Ottawa dataset: This dataset contains pairs of 290×350 SAR images collected in May and August 1997, respectively. These pairs of pictures reflect parts of Ottawa affected by a flood and were provided by Defence Research and Development Canada, Ottawa. Examples are shown in Fig. 8(a) and Fig. 8(b). The ground truth is shown in Fig. 8(c).

Red River dataset: This dataset contains two 512×512 ERS-2 SAR images, which were obtained on August 24, 1996 and August 14, 1999, respectively. Examples are shown in Fig. 9(a) and Fig. 9(b). The ground truth is shown in Fig. 9(c).

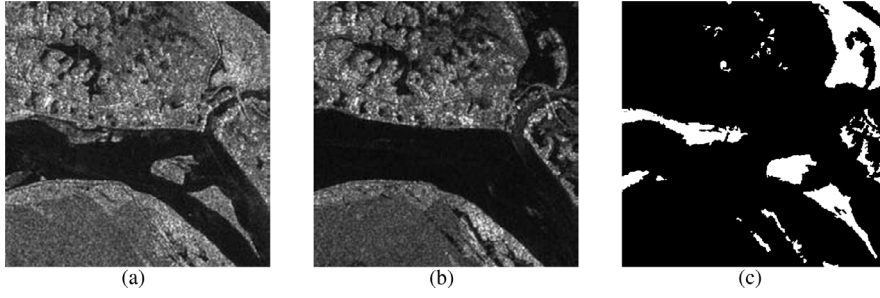


Fig. 9. Multitemporal SAR images of the Red River. (a) image acquired in August 1996. (b) image acquired in August 1999. (c) ground truth.

4.2. Evaluation criteria

Change detection finally outputs a binary image, where white and black represent the changed and unchanged pixels, respectively. We need to compare and analyze the output binary image with our reference image to evaluate our detection result. We can use the relevant guidelines in Rosin and Ioannidis (2003). True negative (TN) represents the correct detection of the number of pixels belonging to the unchanged category; true positive (TP) represents the correct detection of the number of pixels belonging to the changed category; false positive (FP) represents the number of pixels originally belonging to the unchanged class but detected as the changed class, referred to as the number of false detections; and false negative (FN) represents the number of pixels that originally belonged to the changed category but were detected as unchanged category, referred to as the number of missed detections. The overall error (OE) is defined as the sum of the numbers of false and missed detections, which is:

$$OE = FP + FN \quad (20)$$

The percentage of correct classification (PCC) calculation formula is as follows:

$$PCC = \frac{TP + TN}{TP + TN + FP + FN} \quad (21)$$

It is also necessary to calculate the kappa coefficient (KC), a parameter that can more accurately measure the classification accuracy (Brennan & Prediger, 1981). The higher kappa is, the higher the accuracy. Its original calculation formula is as follows:

$$KC = \frac{PCC - PRE}{1 - PRE} \quad (22)$$

PRE represents the proportion of expected agreement:

$$PRE = \frac{(TP + FP) \cdot (TP + FN)}{(TP + TN + FP + FN)^2} + \frac{(FN + TN) \cdot (FP + TN)}{(TP + TN + FP + FN)^2} \quad (23)$$

5. Experimental results

The system environment is as follows: CPU: Intel Core i7-8750H @ 2.20 GHz, GPU: NVIDIA GeForce GTX 1060, RAM: 8.00 GB, Windows 10 Pro (64 b), PyTorch framework (version: 1.6.0), and interpreter: Python 3.8.

5.1. Analysis of the parameters

5.1.1. The neighborhood size

In the proposed method, the neighborhood size of the input patch is an important parameter. Various sized patches contain different spatial information, affecting the central pixel's output label through nonlinear transformation. In other words, the multilevel features learned by the network are directly affected by the information in the input patches. Therefore, the change detection result of multitemporal SAR images is closely related to the neighborhood size.

We conducted experiments on 3×3 , 5×5 , 7×7 , 9×9 , and 11×11 patches to observe the influence of the neighborhood size on the change detection results. Figs. 10(a) and 10(b) show that when the neighborhood size is 3, the OE is larger and the KC is smaller because the neighborhood size is too small, and the center pixel is susceptible to noise interference. When the neighborhood size is 5, the OE and KC are optimal. However, as the neighborhood size gradually increases, the two indicators gradually worsen. The reason is that the larger size of the neighborhood causes the adjacent patches to contain too much identical information, which makes the adjacent pixels be judged as the same category, resulting in the misclassification of the pixels at the boundary. Therefore, the proposed method uses 5×5 as the patch size.

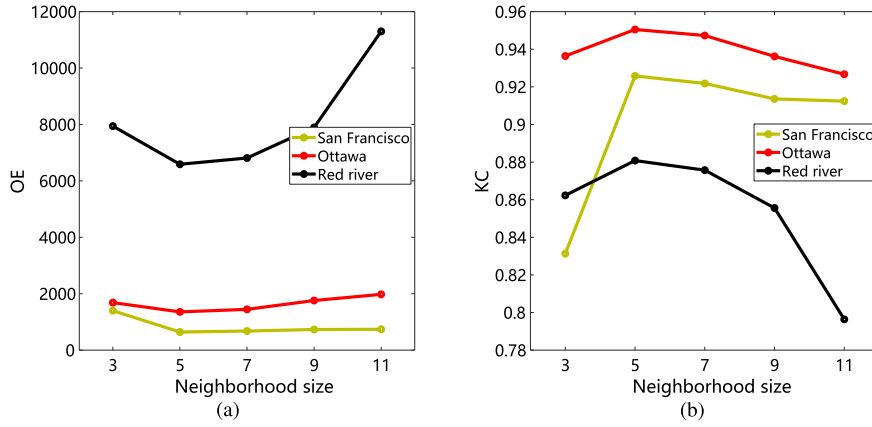


Fig. 10. The influence of the neighborhood size.

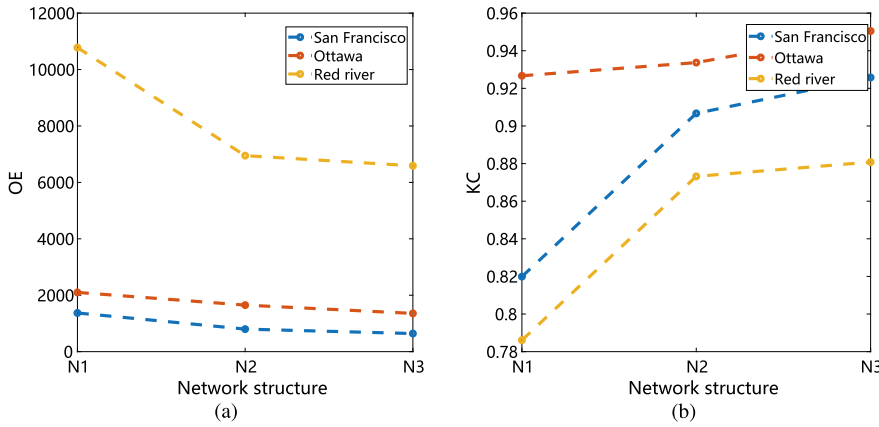


Fig. 11. The influence of the network structure.

5.1.2. Network structure

The network structure is the core of the deep learning algorithm. A shallow network cannot capture complex structural information, and it is easy to cause underfitting. In contrast, the deep structure can obtain high-level context semantic information, but the increase in the parameter space and computational complexity leads to overfitting.

In this article, 25-20-2 (N1), 25-20-15-2 (N2), and 25-20-15-10-2 (N3) structures are used to test the change detection results. The visualization in Fig. 11 shows that the OE decreases and the KC gradually increases as the network layer increases. The better result is because the deep network can learn the complex features of SAR images. The number of neurons in the hidden layer is obtained from practical experience. In addition, the size of the input patch is not suitable for deeper networks, so our method uses N3 as the algorithm structure.

5.1.3. The number of epochs of fine tuning

The number of epochs of fine tuning is another important parameter in the experiment. It affects the effect of the fine tuning of the entire network and the classifier performance. Too few epochs cannot train high-performance classifiers while too many epochs will increase the time complexity and even cause overfitting. Fig. 12 shows the effect of the number of epochs on the change detection results. We can intuitively see that when the number of epochs is less than 15, the detection accuracy is poor. When the number of epochs is greater than 25, OE and KC change slowly or even remain unchanged, indicating that the network performance has been stable. Therefore, the number of epochs is set to 25 in this research.

5.2. Comparison with other methods

The experiment in this article compares five methods: the AE (Hinton & Salakhutdinov, 2006), the VAE (Kingma & Welling, 2014), the DCCNN (Liu et al., 2017), the convolutional-wavelet neural network (CWNN) (Gao et al., 2019) and the dual-domain network (DDNet) (Qu et al., 2021).

In this paper, visualization and quantitative results are used to analyze and compare the models' performances. In the quantitative results, the optimal index data are highlighted in bold.

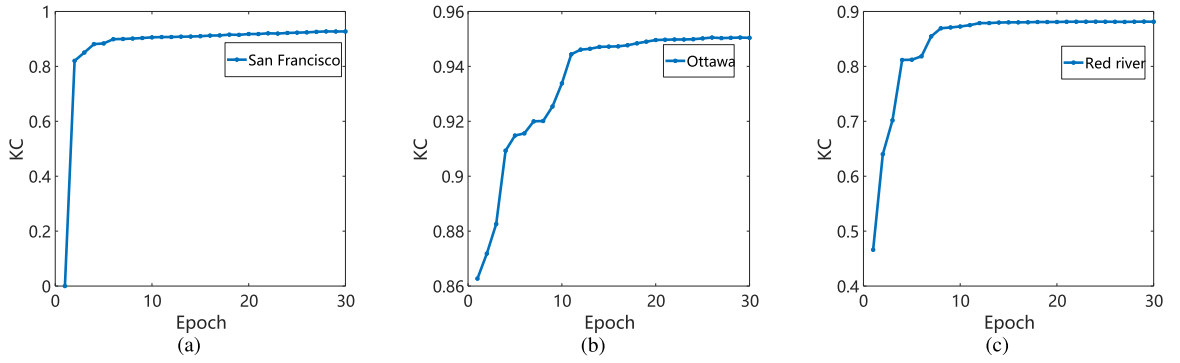


Fig. 12. Relationship between change detection results and the number of epochs of fine tuning. (a) San Francisco dataset. (b) Ottawa dataset. (c) Red River dataset.

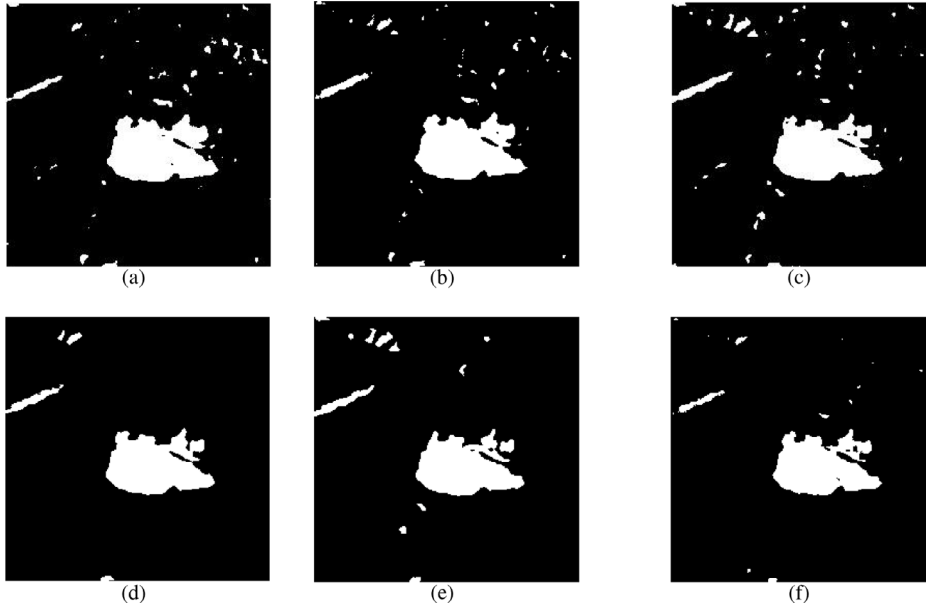


Fig. 13. Change maps on the San Francisco dataset for the following methods: (a) AE, (b) VAE, (c) DCCNN, (d) CWNN, (e) DDNet, and (f) Proposed method.

Table 1

Change detection results on the San Francisco dataset.

Method	FP	FN	OE	PCC	KC
AE	1288	127	1415	0.9784	0.8543
VAE	1016	110	1126	0.9828	0.8812
DCCNN	1169	62	1231	0.9812	0.8724
CWNN	455	185	640	0.9902	0.9283
DDNet	661	233	894	0.9864	0.9014
Proposed	296	346	642	0.9902	0.9258

5.2.1. Results on the San Francisco dataset

Fig. 13 and Table 1 show the visualization and statistical results of the San Francisco dataset. There are many scattered points in Figs. 13(a)–(c), which correspond to the larger FP values of the AE, VAE, and DCCNN methods in Table 1. This shows that these three methods have limited feature expression abilities. In Figs. 13(c)–(e), there are fewer scattered points that are closer to the ground truth. The index data in Table 1 present the effect of each method more clearly. We find that the evaluation index of the VAE algorithm is better than that of the AE, which indicates that the feature learning ability of the VAE is stronger. The results of the proposed method are better than those of the VAE, which also explains why the twin structure can recognize the change information. In addition, this method is not optimal, but its performance is close to that of the CWNN algorithm. For example, the difference in the OE is only 2, the error range of the PCC is $1e-5$, and the KC is only 0.3% smaller than that of the CWNN.

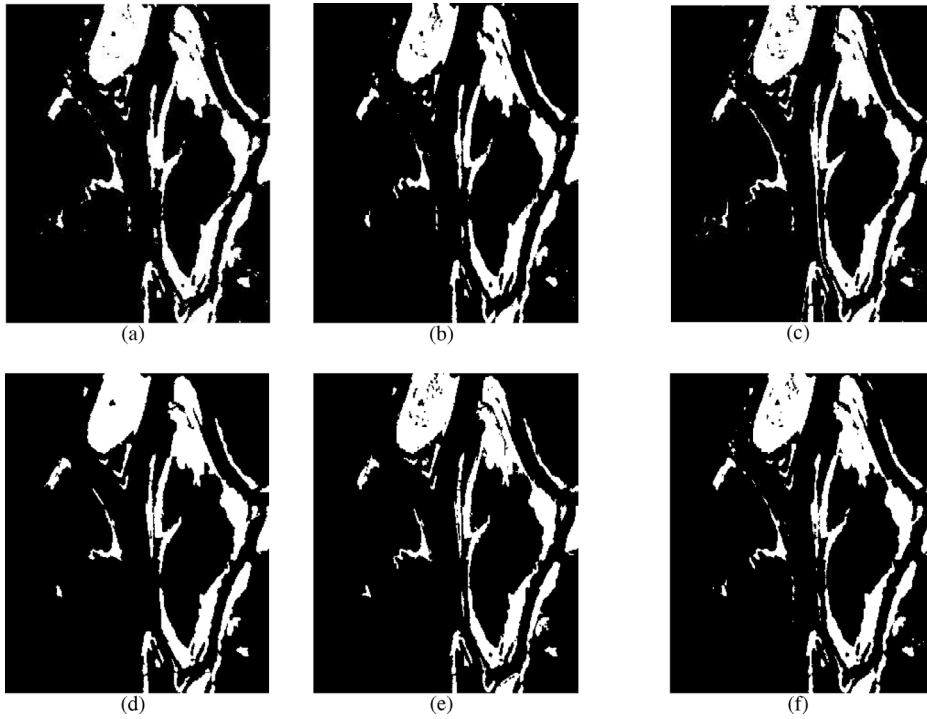


Fig. 14. Change maps on the Ottawa dataset for the following methods: (a) AE, (b) VAE, (c) DCCNN, (d) CWNN, (e) DDNet, and (f) Proposed method.

Table 2
Change detection results on the Ottawa dataset.

Method	FP	FN	OE	PCC	KC
AE	2211	236	2447	0.9759	0.9137
VAE	961	835	1796	0.9823	0.9337
DCCNN	1696	414	2100	0.9793	0.9245
CWNN	1233	484	1717	0.9831	0.9376
DDNet	570	1277	1847	0.9818	0.9304
Proposed	931	425	1356	0.9866	0.9505

5.2.2. Results on the Ottawa dataset

Fig. 14 is the visualization of the Ottawa dataset change detection results. There is some noise at the boundary junction of Fig. 14(a) and Fig. 14(c). Because wavelet denoising mistakenly judges the change point as noise, there is a large wrong area above the middle of Fig. 14(d); therefore, the FPs of the AE, DCCNN, and CWNN in Table 2 are larger. There are many missed pixels in Fig. 14(e), corresponding to the larger FN of DDNet in Table 2. The variation diagram obtained by the proposed method is closer to the reference diagram. The data in Table 2 show that the proposed method is optimal except for the FP and FN, which proves the effectiveness of the proposed method.

5.2.3. Results on the Red River dataset

This dataset has a complex textural structure. In Fig. 9, there are very complex landforms around the river, and its change area is not continuous. There are more local point changes, which will be confused with the speckle noise of the image itself, causing detection difficulties. Fig. 15 and Table 3 show the experimental results of the six methods on the Red River dataset. In Figs. 15(a) and (c), the AE algorithm results in a large number of missing areas (FN = 12227 in Table 3), and the DCCNN method has the most wrongly detected pixels (FP = 12704 in Table 3). There are also many scattered points (false positives) in Fig. 15(b). Due to noise suppression in the frequency domain, there are few scattered points in Fig. 15(e) and Fig. 15(f), but many pixels belonging to the change category are omitted, which corresponds to the larger FN of the CWNN and DDNet in Table 3. The data in Table 3 show that the proposed method is optimal in three important indicators (OE, PCC, and KC). This shows that our method has good performance on complex datasets.

5.3. Robustness analysis

In this subsection, the robustness of the proposed method is verified on a large dataset. This dataset represents a large section (1033 × 715 pixels) of two SAR images acquired by RADARSAT-2 over the region of the Yellow River Estuary of China in June

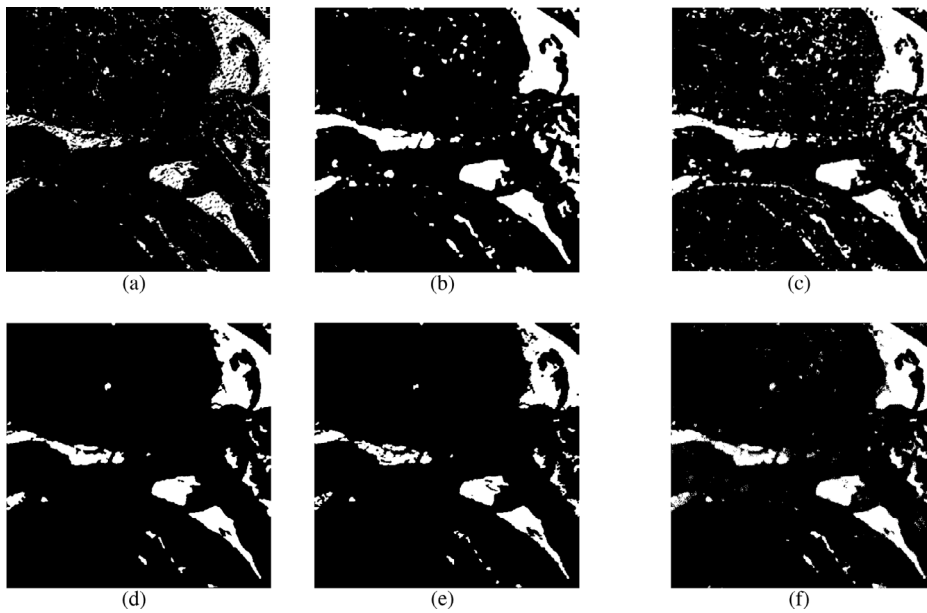


Fig. 15. Change maps on the Red River dataset for the following methods: (a) AE, (b) VAE, (c) DCCNN, (d) CWNN, (e) DDNet, and (f) Proposed method.

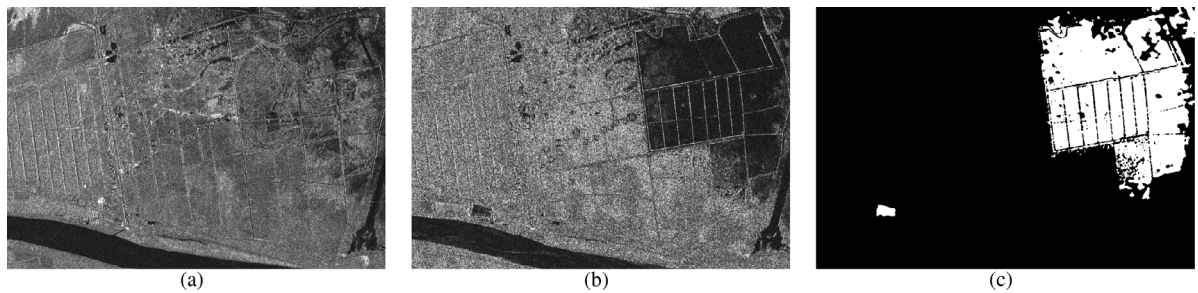


Fig. 16. The large size dataset that represents most regions of the Yellow River. (a) Image acquired in June 2008. (b) Image acquired in June 2009. (c) Ground truth.

Table 3

Change detection results on the Red River dataset.

Method	FP	FN	OE	PCC	KC
AE	2444	12227	14671	0.9440	0.7082
VAE	7193	1192	8331	0.9682	0.8654
DCCNN	12704	718	13422	0.9488	0.7981
CWNN	299	7144	7443	0.9716	0.8584
DDNet	269	8724	8993	0.9657	0.8249
Proposed	1683	4906	6549	0.9748	0.8808

2008 and June 2009. The images to be detected are shown in Fig. 16(a) and Fig. 16(b), and Fig. 16(c) is the ground truth. Since the two original images are one-look images and four-look images, respectively, the influence of speckle noise in the image obtained in 2009 is much greater than that in 2008.

Fig. 17 and Table 4 respectively show the visualization and quantification results of several methods on the Yellow River dataset. The complex dataset makes it difficult to improve the change detection accuracy. There are many scatter points in Fig. 17(a), Fig. 17(b), and Fig. 17(e), which correspond to the higher FP of the AE, VAE, and DDNet methods in Table 4. Figs. 17(e)–(f) show many missing pixels in the white change area, corresponding to the larger FN of the CWNN, the DDNet, and our method in Table 4. The quantitative data in Table 4 show that the proposed method is optimal except for FN, which shows that our algorithm has some advantages in handling large-scale data and reflects the robustness of our algorithm.

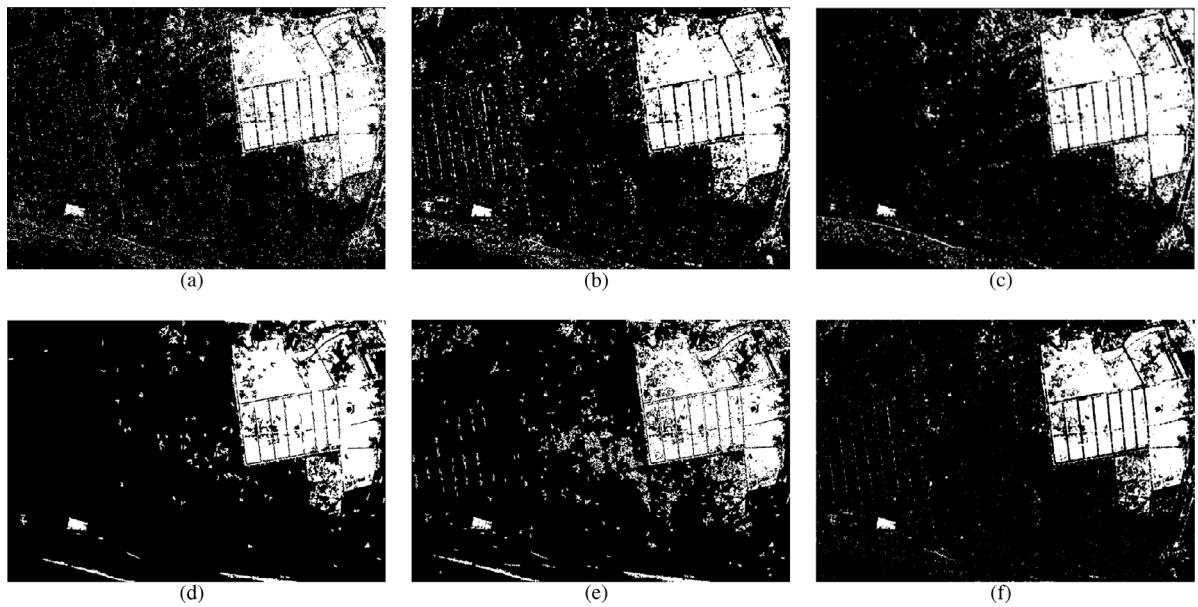


Fig. 17. Change maps on the Yellow River dataset for the following methods: (a) AE, (b) VAE, (c) DCCNN, (d) CWNN, (e) DDNet, and (f) Proposed method.

Table 4

Change detection results on the Yellow River dataset.

Method	FP	FN	OE	PCC	KC
AE	35801	8014	43815	0.9407	0.8116
VAE	34463	7951	42414	0.9426	0.8169
DCCNN	34283	5591	39874	0.9460	0.8248
CWNN	20867	15985	36852	0.9501	0.8249
DDNet	32835	27179	60014	0.9187	0.7249
Proposed	10618	20380	30998	0.9580	0.8511

Table 5

The average training time and testing time.

Method	AE	VAE	DCCNN	CWNN	DDNet	Proposed
Training	36 s	38 s	29 s	38 min	210 s	30 s
Testing	3 s	4 s	6 s	42 s	15 s	3 s

5.4. Time complexity evaluation

The above analysis shows the effectiveness and robustness of the proposed model. In machine learning, the time complexity is an important index to measure an algorithm. Here, we compare the average time consumption of the six methods on four datasets.

Table 5 shows the six comparison methods' average training time and test time on four real datasets. The average training time and test time of the CWNN model were 38 min and 42 s, respectively. The main reason is that supervised training requires more labeled training samples (5000 real samples + 5000 virtual samples). The DDNet uses 7000 supervised samples in the training process, so the computational efficiency is low. Semisupervised training can obtain better results with fewer supervised samples. In addition, the unsupervised feature learning process initializes the parameters, making the fine-tuning phase of the network consume less training time.

The method proposed in this paper, benefiting from the semisupervised learning strategy and the twin network structure of parameter sharing, achieves competitive time complexity. The average training time is only slightly higher than that of the DCCNN, and the test time consumption is less. The results of a comprehensive comparison show that the model and semisupervised learning strategy proposed in this paper achieve competitive change detection accuracy and have advantages in time complexity.

6. Conclusion

In this paper, we proposed a siamese variational autoencoder network to detect changes in SAR images. First, the proposed method uses the log-ratio operator and FCM clustering algorithm to generate pseudolabels and selects high-quality samples according to the sampling strategy. Second, we used the unsupervised method to train the VAE feature extractor layerwise and initially

complete the initialization of network parameters. Next, the labeled samples train the classifier and fine-tune the entire network. Finally, the proposed network classifies the original image to avoid the DI generation process.

The experimental results on four real datasets show that the proposed method is competitive in change detection accuracy and time complexity compared with state-of-the-art technology.

CRedit authorship contribution statement

Guangwei Zhao: Conceptualization, Methodology, Software, Data curation, Writing - Original draft preparation, Visualization, Investigation. **Yaxin Peng:** Supervision, Conceptualization, Methodology, Resources, Validation, Writing - Reviewing and Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This work was supported by the Shanghai Science and Technology Innovation Action Plan (18441909000) and the Capacity Construction Project of Local Universities in Shanghai (no. 18010500600). No funding bodies had any role in the study design, data collection and analysis, decision to publish, or preparation of the manuscript. All authors have read and agreed to the published version of the manuscript.

Data availability

The authors were not given permission to share the data.

References

- Bazi, Y., Bruzzone, L., & Melgani, F. (2005). An unsupervised approach based on the generalized Gaussian model to automatic change detection in multitemporal SAR images. *IEEE Transactions on Geoscience and Remote Sensing*, 43(4), 874–887.
- Brennan, R. L., & Prediger, D. J. (1981). Coefficient kappa: Some uses, misuses, and alternatives. *Educational and Psychological Measurement*, 41(3), 687–699.
- Bromley, J., Bentz, J. W., Bottou, L., Guyon, I., Lecun, Y., Moore, C., et al. (1993). Signature verification using a siamese time delay neural network. *International Journal of Pattern Recognition and Artificial Intelligence*, 7(4), 669–688.
- Bruzzone, L., & Prieto, D. F. (2000). Automatic analysis of the difference image for unsupervised change detection. *IEEE Transactions on Geoscience and Remote Sensing*, 38, 1171–1182.
- Bruzzone, L., & Prieto, D. F. (2002). An adaptive semiparametric and contextbased approach to unsupervised change detection in multitemporal remote-sensing images. *IEEE Transactions on Image Processing*, 11(4), 452–466.
- Chen, H., Jiao, L., Liang, M., Liu, F., Yang, S., & Hou, B. (2019). Fast unsupervised deep fusion network for change detection of multitemporal SAR images. *Neurocomputing*, 332(7), 56–70.
- Dalca, A., Balakrishnan, G., Guttag, J., & Sabuncu, M. R. (2018). Unsupervised learning for fast probabilistic diffeomorphic registration. In *MICCAI 2018: medical image computing and computer assisted intervention* (pp. 729–738).
- Fernandez-Prieto, D., & Marconcini, M. (2011). A novel partially supervised approach to targeted change detection. *IEEE Transactions on Geoscience and Remote Sensing*, 49(12), 5016–5038.
- Gao, F., Wang, X., Gao, Y., Dong, J., & Wang, S. (2019). Sea ice change detection in SAR images based on convolutional-wavelet neural networks. *IEEE Geoscience and Remote Sensing Letters*, 16(8), 1240–1244.
- Ghosh, A., Mishra, N., & Ghosh, S. (2011). Fuzzy clustering algorithms for unsupervised change detection in remote sensing images. *Information Sciences*, 181(4), 699–715.
- Glorot, X., Antoine, B., & Yoshua, B. (2011). Deep sparse rectifier neural networks. *Journal of Machine Learning Research*, 15, 315–323.
- Gong, M., Cao, Y., & Wu, Q. (2012). A neighborhood-based ratio approach for change detection in SAR images. *IEEE Geoscience and Remote Sensing Letters*, 9(2), 307–311.
- Gong, M., Li, Y., Jiao, L., Jia, M., & Su, L. (2014). SAR change detection based on intensity and texture changes. *ISPRS Journal of Photogrammetry and Remote Sensing*, 93(1), 123–135.
- Gong, M., Zhan, T., Zhang, P., & Miao, Q. (2017). Superpixel-based difference representation learning for change detection in multispectral remote sensing images. *IEEE Transactions on Geoscience and Remote Sensing*, 55(5), 2658–2673.
- Gong, M., Zhao, J., Liu, J., Miao, Q., & Jiao, L. (2016). Change detection in synthetic aperture radar images based on deep neural networks. *IEEE Transactions on Neural Networks and Learning Systems*, 27(1), 125–138.
- Hinton, G. E., & Salakhutdinov, R. R. (2006). Reducing the dimensionality of data with neural networks. *Science*, 313(5786), 504–507.
- Hou, B., Wei, Q., Zheng, Y., & Wang, S. (2014). Unsupervised change detection in SAR image based on Gauss-log ratio image fusion and compressed projection. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 7(8), 3297–3317.
- Hu, M., Zhong, Y., Xie, S., Lv, H., & Lv, Z. (2021). Fuzzy system based medical image processing for brain disease prediction. *Frontiers in Neuroscience*, <http://dx.doi.org/10.3389/fnins.2021.714318>.
- Inglada, J., & Mercier, G. (2007). A new statistical similarity measure for change detection in multitemporal SAR images and its extension to multiscale change analysis. *IEEE Transactions on Geoscience and Remote Sensing*, 45(5), 1432–1445.
- Kingma, D. P., & Welling, M. (2014). Auto-encoding variational Bayes. In *International conference on learning representations* (pp. 1–14).
- Kittler, J., & Illingworth, J. (1986). Minimum error thresholding. *Pattern Recognition*, 19(1), 41–47.
- Li, Y., Martinis, S., Plank, S., & Ludwig, R. (2018). An automatic change detection approach for rapid flood mapping in Sentinel-1 SAR data. *International Journal of Applied Earth Observation and Geoinformation*, 73, 123–135.
- Li, Y., Peng, C., Chen, C., Jiao, L., Zhou, L., & Shang, R. (2019). A deep learning method for change detection in synthetic aperture radar images. *IEEE Transactions on Geoscience and Remote Sensing*, 57(8), 5751–5763.

- Li, H., Yang, G., Yang, W., Du, Q., & Emery, W. J. (2020). Deep nonsmooth nonnegative matrix factorization network with semi-supervised learning for SAR image change detection. *ISPRS Journal of Photogrammetry and Remote Sensing*, 160, 167–179.
- Liu, S., Bruzzone, L., Bovolo, F., & Du, P. (2015). Hierarchical unsupervised change detection in multitemporal hyperspectral images. *IEEE Transactions on Geoscience and Remote Sensing*, 53(1), 244–260.
- Liu, J., Gong, M., Qin, K., & Zhang, P. (2018). A deep convolutional coupling network for change detection based on heterogeneous optical and radar images. *IEEE Transactions on Neural Networks and Learning Systems*, 29(3), 545–559.
- Liu, T., Li, Y., Cao, Y., & Shen, Q. (2017). Change detection in multitemporal synthetic aperture radar images using dual-channel convolutional neural network. *Journal of Applied Remote Sensing*, 11(4), 1–25.
- Liu, R., Wang, R., Huang, J., Li, J., & Jiao, L. (2020). Change detection in SAR images using multiobjective optimization and ensemble strategy. *IEEE Geoscience and Remote Sensing Letters*, 1–5. <http://dx.doi.org/10.1109/LGRS.2020.3005140>.
- Ma, J., Gong, M., & Zhou, Z. (2012). Wavelet fusion on ratio images for change detection in SAR images. *IEEE Geoscience and Remote Sensing Letters*, 9(6), 1122–1126.
- Meng, H., Gao, L., Lai, Y., & Manocha, D. (2019). VV-Net: Voxel VAE net with group convolutions for point cloud segmentation. In *2019 IEEE/CVF international conference on computer vision (ICCV)* (pp. 8499–8507).
- Mishra, N. S., Ghosh, S., & Ghosh, A. (2012). Fuzzy clustering algorithms incorporating local information for change detection in remotely sensed images. *Applied Soft Computing*, 12(8), 2683–2692.
- Planinsic, P., & Gleich, D. (2018). Temporal change detection in SAR images using log cumulants and stacked autoencoder. *IEEE Geoscience and Remote Sensing Letters*, 15(2), 297–301.
- Prasad, V., Das, D., & Bhowmick, B. (2020). Variational clustering: leveraging variational autoencoders for image clustering. In *2020 international joint conference on neural networks (IJCNN)* (pp. 1–10).
- Qu, X., Gao, F., Dong, J., Du, Q., & Li, H. (2021). Change detection in synthetic aperture radar images using a dual-domain network. *IEEE Geoscience and Remote Sensing Letters*, <http://dx.doi.org/10.1109/LGRS.2021.3073900>.
- Rosin, P. L., & Ioannidis, E. (2003). Evaluation of global image thresholding for change detection. *Pattern Recognition Letters*, 24(14), 2345–2356.
- Shang, R., Yuan, Y., Jiao, L., Meng, Y., & Ghalamzan, A. (2018). A self-paced learning algorithm for change detection in synthetic aperture radar images. *Signal Processing*, 142, 375–387.
- Shi, W., Zhang, M., Zhang, R., Chen, S., & Zhao, Z. (2020). Change detection based on artificial intelligence: State-of-the-art and challenges. *Remote Sensing*, 12(10), 1688.
- Su, L., Gong, M., Sun, B., & Jiao, L. (2014). Unsupervised change detection in SAR images based on locally fitting model and semi-EM algorithm. *International Journal of Remote Sensing*, 35(2), 621–650.
- Van Engelen, J. E., & Hoos, H. H. (2020). A survey on semi-supervised learning. *Machine Learning*, 109(2), 373–440.
- Wang, S., Jiao, L., & Yang, S. (2016). SAR images change detection based on spatial coding and nonlocal similarity pooling. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 9(8), 3452–3466.
- Zhang, C., Yue, P., Tapete, D., Jiang, L., Shanguan, B., Huang, L., et al. (2020). A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images. *ISPRS Journal of Photogrammetry and Remote Sensing*, 166, 183–200.
- Zheng, Y., Jiao, L., Liu, H., Zhang, X., Hou, B., & Wang, S. (2017). Unsupervised saliency-guided SAR image change detection. *Pattern Recognition*, 61, 309–326.